HAMZA EL BYAD

MACHINE LEARNING ENGINEER · APPLIED AI · SCIENTIFIC COMPUTING

I build ML systems that work beyond the notebook.

PhD in Mathematics & Computer Science, with experience developing RAG applications, industrial predictive models and statistical software for scientific research. I bring together model evaluation, reliable software, and release engineering.

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WHY MY EXPERIENCE IS RELEVANT

Applied AI engineering. Designed and validated stateful retrieval workflows that combine LLM interfaces with deterministic application rules.

Industrial machine learning. Built spectral-processing and predictive-modeling experiments with attention to grouped evaluation and data leakage.

Scientific research. Applied statistical computing to astrophysical time series and contributed to peer-reviewed research.

SELECTED WORK

CustomQuoteAI: Released private internal-estimation MVP with three successful CI jobs and 13 documented browser acceptance steps.

NMR property prediction: Reported experimental API-gravity model with R^2 0.963 and RMSE 1.312; external validation outstanding.

Astrophysics: First-author publication and scientific software work in time-series analysis.

All project descriptions are generalized. Private source code and catalog data are not reproduced.

01 / APPLIED AI ENGINEERING

CustomQuoteAI

What problem did I solve?

Complex configuration catalogs are hard to search by hand. One selected option can include multiple components, units, signed adjustments and estimated costs. A chatbot alone is insufficient if retrieval returns incompatible records or invents financial values.

What did I build?

A conversational internal-estimation assistant combining FAISS semantic retrieval, deterministic exact-code lookup, context-aware filtering, and LangGraph conversation state. Users search, choose component groups, add or remove options and review cost evidence.

Engineering decisions

Exact option identifiers are handled deterministically. Component quantities and source totals stay intact, including zeros and credits. Estimated costs are never promoted into customer selling prices; issuance and priced CSV output remain blocked.

Evidence, not inflated impact

Version: v1.0.0, private annotated release through GitFlow. Quality: backend, frontend and secret-scanning jobs passed. Acceptance: 13 recorded browser steps passed.

Current limitation: a later pilot exposed an unexpected multi-gigabyte embedding-model download and first-search delay. Model/cache root cause was not established. The release should not be described as proven production-scale deployment.

02 / INDUSTRIAL MACHINE LEARNING

NMR Property Prediction

Machine learning problem

Estimate crude-oil properties from proton NMR measurements through a reproducible preprocessing and regression workflow. The principal challenge is producing trustworthy estimates from a small sample set with repeated measurements.

Pipeline and evaluation

Signal preparation: windowing, Fourier transformation, phase and baseline correction, spectral alignment and normalization. Modeling: PCR and PLS regression with consistent preprocessing. Evaluation: grouped validation to avoid placing replicate spectra from the same physical sample in training and validation simultaneously.

Reported experimental result

The best project-reported API-gravity configuration achieved $R^2 = 0.963$ and RMSE = 1.312. This is an internal experimental metric, not evidence of independent generalization to new batches.

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03 / SCIENTIFIC RESEARCH

Astrophysics & Scientific Computing

Developed statistical methods and scientific Python workflows for noisy high-energy astrophysics observations, including frequency-domain analysis, uncertainty estimation and weak-signal detection.

The work includes a first-author Astrophysical Journal paper and co-authored scientific-computing publications. See the research page for publication titles and DOI links.

ENGINEERING & RESEARCH PROFILE

Methods, tools and evidence

Core technical strengths

Machine learning and statistics: Python, PyTorch, scikit-learn, regression, signal processing, time-series analysis and validation design.

Applied AI: retrieval-augmented generation, FAISS, LangGraph, FastAPI, Chainlit and deterministic application logic.

Engineering: GitFlow, pytest, GitHub Actions, Docker, TypeScript, vulnerability remediation, regression testing and release tagging.

Selected research

First-author article in The Astrophysical Journal 989(2), 202 (2025), alongside other co-authored astrophysics and scientific-software publications. Full bibliographic links appear on the portfolio website.

What the evidence does—and does not—prove

Successful automated tests and versioned releases demonstrate engineering discipline. Experimental regression metrics need external validation for claims of generalization. A Git release does not guarantee that model artifacts, caches and inference infrastructure are ready in a fresh runtime.

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Code and confidential company datasets for the internal assistant remain private. No confidential data or financial records are included in this portfolio.